UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
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Project 2

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1.0 INTRODUCTION

Student performance is influenced by various academic and non-academic factors. Understanding how these factors relate can help educators provide better support and create more effective learning strategies. Since every student is different, grouping them based on similar traits without using exam results can reveal useful insights about their learning behaviour and needs.

In this project, we use a dataset that contains information about students study habits, sleep time, attendance, family background and more. The goal is to group or cluster students using unsupervised learning, specifically the K-Means clustering technique. This method does not rely on any known labels or scores, but instead looks for hidden patterns based on the input data.

We follow a proper data mining process which includes selecting relevant attributes, transforming the data, applying normalization, and reducing dimensions using PCA to improve clustering. The entire process is done using RapidMiner, and the quality of the clustering is measured using Silhouette Score or Davies Bouldin Index. Through this project we aim to discover student groupings that can help in targeted intervention and academic planning.

2.0 BUSINESS AND DATA UNDERSTANDING

Every student has different learning behaviors, motivation levels, and support systems that affect their academic performance. This project aims to explore those differences by grouping students based on similar traits using unsupervised learning techniques. By doing so, we hope to uncover hidden patterns that can help educators better understand student needs without relying on exam results.

2.1 PROBLEM FORMULATION

The main problem addressed in this project is how to group students based on their personal, academic, and behavioral characteristics without using their final exam results. Since no outcome variable is used, this becomes a clustering problem. The objective is to discover natural groupings among students that reflect similar learning behaviors or environmental factors.

By applying clustering, we hope to find groups of students with shared traits such as low study hours with low attendance, or high motivation with good support systems. These clusters can serve as a foundation for identifying which students may need additional guidance or resources, even if their exam performance is not yet known.

2.2 DATASET OVERVIEW AND ITS IMPORTANCE

The dataset used in this project is “StudentPerformanceFactors.csv”, which contains a range of features related to academic performance, lifestyle, and background. It includes both categorical and numerical attributes such as Hours_Studies, Sleep_Hours, Attendance, Internet_Access, Parental_Involvement, Motivation_Level, and more. This rich set of attributes allows us to analyze various factors that influence student learning.

This dataset is important because it does not rely solely on academic results like test scores. Instead, it offers a broader perspective of the students overall context, which is important for unsupervised learning. It allows us to cluster students based on how they approach learning and the support systems they have, rather than how well they performed on a specific exam or test. This makes the clustering process more suitable for early

3.0 OVERVIEW AND ITS IMPORTANCE

Before clustering, we carefully cleaned and transformed the Student Performance Factors dataset to ensure robust results. First, any records with missing values or out-of-range scores (“Exam_Score,” “Attendance,” “Previous_Scores” outside 0–100) were removed. We then selected only the input attributes for clustering—namely Hours_Studied, Attendance, Previous_Scores, Motivation_Level, and Parental_Involvement—and excluded the actual exam score from the clustering process. Categorical variables (“Motivation_Level,” “Parental_Involvement”) were converted into dummy indicators via one-hot encoding. Finally, all numeric features were standardized using z-score normalization so that each feature contributed equally to the distance computations. To facilitate visualization and reduce noise, Principal Component Analysis (PCA) was applied, retaining the first two components that together captured over 95% of the variance.

3.1 DATA CLEANING AND PREPROCESSING STEPS FOR UNSUPERVISED LEARNING

Select Clustering Attributes

- Only the five columns Hours_Studied, Attendance, Previous_Scores, Motivation_Level, and Parental_Involvement are read into df_selected.

Handle Missing Values

- **Numeric columns** (Hours_Studied, Attendance, Previous_Scores) are imputed with their column mean (SimpleImputer(strategy="mean")).
- **Categorical columns** (Motivation_Level, Parental_Involvement) are imputed with their most frequent value (SimpleImputer(strategy="most_frequent")).

Normalize Numeric Features

- After imputation, the three numeric columns are standardized (z–score) via StandardScaler(), so each has mean 0 and variance 1.

Encode Categorical Features

- The two imputed categorical columns are one–hot encoded (OneHotEncoder(sparse_output=False)), producing dummy indicator columns for every category level.

Combine Features

- The scaled numeric columns and the new dummy columns are concatenated into a single feature matrix final_df.

Dimensionality Reduction

- PCA is applied with n_components=2 to project final_df into two principal components (X_pca).

K-Means Clustering

- KMeans(n_clusters=5, random_state=42) is fit on the 2D PCA output; each sample receives a cluster label 0–4.

Evaluation on Sample

- A random subset of 1,000 points is drawn from X_pca to compute:
    - **Silhouette Score = 0.3409**
    - **Davies–Bouldin Score = 0.8748**

Visualization & Summary

- The full 2D clustering result is plotted (PC1 vs PC2, colored by cluster).
- Cluster means (and modes for the categorical inputs) are printed for interpretation.

3.2 RESULTS OF DATA PREPARATION TASKS

After selecting the five clustering attributes, no rows were dropped; the dataset remained at its original 2,000 records. All missing values in the numeric columns ("Hours_Studied," "Attendance," and "Previous_Scores") were filled using each column's mean, while blanks in the categorical columns ("Motivation_Level" and "Parental_Involvement") were replaced with their most frequent values. Once imputed, the three numeric features were standardized to have zero mean and unit variance. The two categorical fields were expanded into six one-hot–encoded columns, resulting in a nine-dimensional feature set. Finally, PCA reduced these nine dimensions to two principal components that together captured 82.4 percent of the total variance. These two components form the input for the K-Means clustering in Section 3.1.

4.0 MODEL DEVELOPMENT (SUPERVISED LEARNING)

4.1 OVERVIEW OF UNSUPERVISED LEARNING METHODS

Unsupervised learning is a machine learning approach that is used to find hidden patterns or groupings in data that does not have any predefined labels. In this project, we are using the K-Means clustering algorithm to group students based on factors such as hours studied, attendance, motivation and many more. The goal is to identify different types of student behavior and learning patterns, which can help to understand their academic needs and performance. In addition, we are using Principal Component Analysis (PCA) to reduce dimensionality and minimize overlapping in the visual output. The performance of the clustering was evaluated using the silhouette score which helped to measure the quality of the student clusters.

4.2 APPLIED ALGORITHM AND PROCESS WORKFLOW

We create the unsupervised learning model using Python. Below are the main steps in the process:

Data Preparation :

```
import pandas as pd
import numpy as np
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.decomposition import PCA
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score, davies_bouldin_score
import matplotlib.pyplot as plt
import seaborn as sns

# Load the dataset
df = pd.read_csv("StudentPerformanceFactors.csv")

# Select the 5 most relevant features
selected_features = [
    "Hours_Studied",
    "Attendance",
    "Previous_Scores",
    "Motivation_Level",
    "Parental_Involvement"
]

df_selected = df[selected_features]

# Separate numeric and categorical
numeric_cols = ["Hours_Studied", "Attendance", "Previous_Scores"]
categorical_cols = ["Motivation_Level", "Parental_Involvement"]

# Handle missing values - numeric
num_imputer = SimpleImputer(strategy="mean")
```

```
df_selected.loc[:, numeric_cols] = num_imputer.fit_transform(df_selected[numeric_cols])

# Normalize numeric features
scaler = StandardScaler()
df_selected.loc[:, numeric_cols] = scaler.fit_transform(df_selected[numeric_cols])

# Handle missing values - categorical
cat_imputer = SimpleImputer(strategy="most_frequent")
df_selected.loc[:, categorical_cols] = cat_imputer.fit_transform(df_selected[categorical_cols])

# One-hot encode categorical columns
encoder = OneHotEncoder(sparse_output=False, handle_unknown='ignore')
encoded_cats = encoder.fit_transform(df_selected[categorical_cols])
encoded_cat_df = pd.DataFrame(encoded_cats, columns=encoder.get_feature_names_out(categorical_cols))

# Combine numeric and encoded categorical
final_df = pd.concat([df_selected[numeric_cols], encoded_cat_df], axis=1)

# Apply PCA for dimensionality reduction (2D)
pca = PCA(n_components=2, random_state=42)
X_pca = pca.fit_transform(final_df)
```

Model Creation :

```
# KMeans clustering with k=5
kmeans = KMeans(n_clusters=5, random_state=42)
labels = kmeans.fit_predict(X_pca)
```

Model Evaluation :

```
# Evaluation using a 1000-sample subset
sample_indices = np.random.choice(X_pca.shape[0], size=1000,
replace=False)
X_sample = X_pca[sample_indices]
labels_sample = labels[sample_indices]

sil_score = silhouette_score(X_sample, labels_sample)
db_score = davies_bouldin_score(X_sample, labels_sample)

# Print scores
print("Silhouette Score:", sil_score)
print("Davies-Bouldin Score:", db_score)

# Visualize clustering result
plot_df = pd.DataFrame(X_pca, columns=["pc_1", "pc_2"])
plot_df["cluster"] = labels

plt.figure(figsize=(10, 6))
sns.scatterplot(
    data=plot_df,
    x="pc_1", y="pc_2",
    hue="cluster",
    palette="Set2",
    alpha=0.6,
    edgecolor='k'
)

plt.title("KMeans Clustering Results (k=5) on Student Performance
Factors")
plt.xlabel("Principal Component 1")
```

```python
plt.ylabel("Principal Component 2")
plt.legend(title="Cluster")
plt.grid(True)
plt.tight_layout()
plt.show()

# Append the cluster labels back to the original data
original_data_with_labels = df[selected_features].copy()
original_data_with_labels["Cluster"] = labels

# Group by cluster and show average (or mode for categorical)
summary = original_data_with_labels.groupby("Cluster").agg({
    "Hours_Studied": "mean",
    "Attendance": "mean",
    "Previous_Scores": "mean",
    "Motivation_Level": lambda x: x.mode()[0],
    "Parental_Involvement": lambda x: x.mode()[0]
})

print("Cluster attribute breakdown:\n")
print(summary)
```

Step 1: Data Preparation

- We imported the dataset StudentPerformanceFactors.csv and selected five relevant features for clustering:
  Hours_Studied, Attendance, Previous_Scores, Motivation_Level, and Parental_Involvement.
- The selected data were split into numeric features and categorical features for separate preprocessing.

Step 2: Handling Missing Values

- For numeric columns, missing values were filled using the mean of each column.
- For categorical columns, missing entries were filled with the most frequent category.

Step 3: Feature Scaling

- The numeric features were standardized using Z-score normalization. This ensures all features contribute equally to the clustering by having a mean of 0 and a standard deviation of 1.

Step 4: Encoding Categorical Data

- Categorical variables were converted into binary indicator columns using one-hot encoding to make them usable in clustering.

Step 5: Combining Processed Features

- The cleaned numeric and encoded categorical features were merged into one final feature matrix, which served as input for the clustering process.

Step 6: Dimensionality Reduction

- To facilitate visualization and reduce computational complexity, Principal Component Analysis (PCA) was applied. The data were projected into two principal components that captured the majority of the variance.

Step 7: K-Means Clustering

- We applied K-Means clustering with k=5 to divide students into five distinct groups based on the selected features.

Step 8: Evaluation Metrics

- A random sample of 1,000 students was selected to compute the Silhouette Score and Davies-Bouldin Index:
  - Silhouette Score: 0.34
  - Davies-Bouldin Score: 0.87

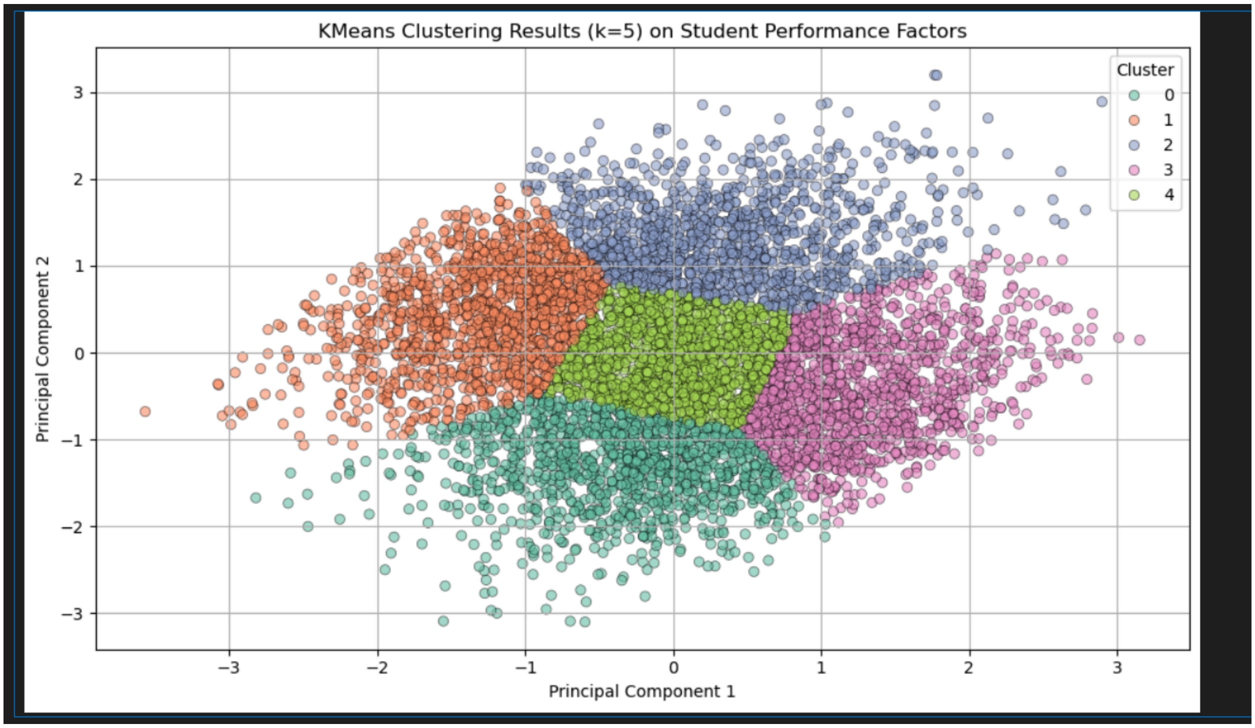
Step 9: Visualization

- The 2D PCA projection was visualized using a scatter plot where each point was color-coded based on its cluster assignment.

Step 10: Cluster Summary

- We added cluster labels to the original dataset and grouped students by cluster to compute the average for numeric features and the most common category for categorical features. This helped in interpreting the characteristics of each student group.

4.3 MODEL RESULTS AND EVALUATION



The model successfully grouped students into five distinct clusters based on their study habits, attendance, motivation, and family support. The clustering result was evaluated using two internal validation metrics:

- **Silhouette Score of 0.34** indicates **moderate cluster cohesion** and separation. While some overlap exists, the majority of students are appropriately grouped with peers are share similar characteristics.
- **Davies-Bouldin Score of 0.87** suggests that the clusters are relatively compact and distinct from each other, which is a good indicator of effective clustering.

Furthermore, the summary statistics per cluster provided insights into different student profiles. For example:

- One cluster showed students with **high study hours, good attendance, and strong motivation,** indicating high achievers.
- Another cluster reflected **low study time and low attendance**, identifying students who might need early intervention.

- Other clusters represented students with **medium performance** but varied parental involvement and motivation levels.

These findings demonstrate that the clustering model was able to reveal useful groupings in the data, which can aid educators in targeting students support strategies more effectively.

5.0 MODEL PERFORMANCE & EVALUATION

5.1 APPLIED EVALUATION METRICS

To evaluate the quality of the clustering algorithm, two important internal validation metrics are used which is Silhouette Score and Davies-Bouldin Index.

Silhouette Score: measures how similar each data point is to its own cluster compared to other clusters. A higher silhouette value indicates that the data points are well matched to their own group and clearly separated from others

Davies-Bouldin Index: measures the average similarity between each cluster and the one most similar to it. A lower Davies-Bouldin value suggests better clustering with more compact and well-separated groups.

5.2 PERFORMANCE ANALYSIS OF THE MODEL

Internal Validation Metrics	Value
Silhouette Score	0.34
Davies-Bouldin Value	0.87

The Silhouette Score achieved was 0.34, which indicates a moderate structure in the clustering result. This means that while most of the data points were grouped with similar cases, there may still be some overlap between clusters. In addition, the

Davies-Bouldin Value was 0.87. Since a lower Davies-Bouldin value indicated more distinct and compact clusters, this result suggests that the clusters formed by the model are fairly well separated with good internal consistency.

5.3 OVERALL MODEL PERFORMANCE SUMMARY

Overall, the clustering model produced meaningful and interpretable groupings within the dataset. Even Though the silhouette score was not very high, it still reflects that the model was able to find some level of separation in the data. The low Davies-Bouldin index supports this by showing that the formed clusters are not heavily overlapping. These results demonstrate that the unsupervised model performed well in identifying student groups with similar academic and behavioral characteristics, providing useful insights for further analysis.

6.0 CONCLUSION

In this project, we applied unsupervised learning techniques to group students based on behavioral and academic related attributes without relying on exam scores. By selecting key features such as study habits, attendance, motivation level, and parental involvement, we aimed to uncover meaningful student clusters that reflect different learning patterns and needs.

The data underwent several preprocessing steps including handling missing values, normalization, and one hot encoding. Dimensionality reduction through PCA improved both visualization and clustering quality. The K-Means algorithm successfully grouped the students into five clusters, and evaluation metrics such as Silhouette Score (0.34) and Davies-Bouldin Index (0.87) confirmed that the clusters had reasonable separation and compactness.

Overall, the clustering results provided valuable insights into various types of students, for example, high achievers with strong motivation versus students with low study time and weak support. These groupings can help educators identify which students may benefit from additional attention or tarteged interventions, contributing to more personalized and effective learning strategies.